

Siddartha Reddy Munugala

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PROFESSIONAL SUMMARY

Software Engineer specializing in full-stack development, AI-powered applications, and scalable backend systems. Built production web applications using TypeScript, Next.js, Python, PostgreSQL, and OpenAI APIs, with experience developing real-time communication platforms, developer tools, and machine learning applications. Passionate about building reliable, high-performance software from idea to deployment.

Core Technologies: Python • TypeScript • React • Next.js • Node.js • PostgreSQL • Docker • Git • CI/CD

EDUCATION

University of South Dakota

Vermillion, SD • Expected May 2027

B.S. Computer Science — Minor in Innovation & Entrepreneurship

Relevant Coursework: Data Structures, Algorithms, Systems Programming, Database Systems, Software Engineering, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python • TypeScript • JavaScript • SQL • Linux • OOP

Frontend: React • Next.js • Tailwind CSS • CodeMirror 6

Backend: Node.js • RESTful APIs • WebSockets • WebRTC • FastAPI • API Design • Authentication • Authorization

Databases: PostgreSQL • Supabase • Firebase • Redis • Neon • Database Design • Caching

AI / Machine Learning: OpenAI API • Claude API • Gemini API • LangChain • PyTorch • XGBoost • SHAP • Whisper

Cloud & DevOps: Docker • Git • GitHub Actions • Vercel • CI/CD • Caching • Monitoring • Observability

Developer Tools: FFmpeg • tree-sitter • NetworkX • Agile

EXPERIENCE

Co-Founder & Lead Engineer — VibeIn / ArenaHQ

Jan 2025 – Present

vibein.us • arenahq.us

- Built and launched two full-stack web applications using Next.js, Supabase, and WebRTC, growing the platforms to over 1,600 registered users.
- Resolved critical production issues involving Supabase RLS policies, API migrations, and real-time database performance to improve application reliability.
- Implemented real-time messaging and secure peer-to-peer video communication using WebSockets and WebRTC.
- Automated deployments using GitHub Actions and Vercel, reducing manual releases and improving deployment consistency.

Production Operations Staff — USD Athletics

Aug 2025 – Present

University of South Dakota • Part-Time On-Campus

- Operated live-event audio systems and digital display infrastructure for NCAA Division I broadcasts under zero-tolerance real-time constraints, ensuring seamless production for televised games.

IT Coordinator & Publicity Chair — NSLS Executive Board

Sep 2024 – Present

University of South Dakota

- Automated member onboarding pipelines and communication workflows using Python and custom dashboards, reducing manual administrative overhead by 50% for a 100+ member organization.

PROJECTS

EKO — AI Background Music & Voice Dubbing System

2026 – Present

- Developed a three-mode AI audio platform integrating speech translation, video scene-energy music matching, and multilingual voice dubbing.
- Built the system architecture using Next.js, FastAPI, Neon Postgres, and Demucs stem-separation algorithms for real-time audio isolation.
- Reduced inference server memory usage from 1.8 GB to 400 MB by moving model execution to the Hugging Face Inference API, eliminating container crashes.
- Deployed the application to over 800 users organically and drafted a technical system architecture preprint.

PRISM — PR Risk Intelligence & Structural Modeling

2026

- Engineered an automated machine learning pipeline analyzing AST code complexity (via tree-sitter) and dependency graphs (via NetworkX) to predict post-merge software regression risks.
- Trained and validated an XGBoost classifier with SHAP-based explainability, achieving robust generalization across diverse code repositories.

LEADERSHIP & ACTIVITIES

- Designed and taught AI workshops for 50+ students covering machine learning fundamentals, ethics, and practical applications.
- Mentored local youth in offline community programs on foundational business, customer discovery, and communication skills.
- Authored a 10-week strategic operational plan and built the chapter administration portal for USD INFORMS.
- Coordinated campus tech events as IT Coordinator and Publicity Chair for the NSLS Executive Board.